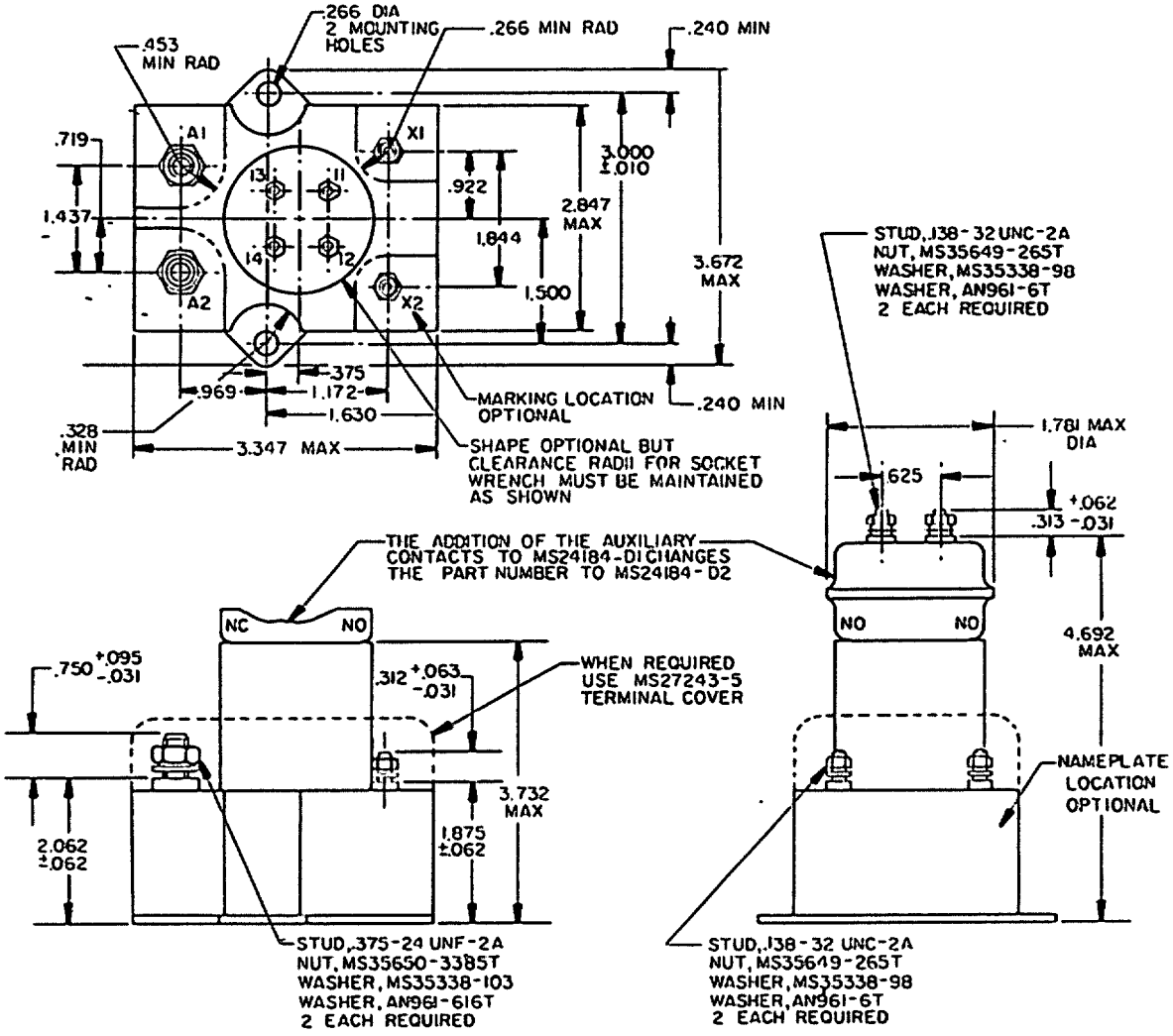
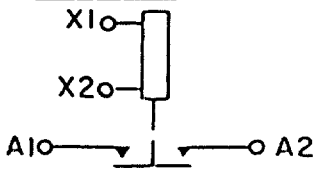
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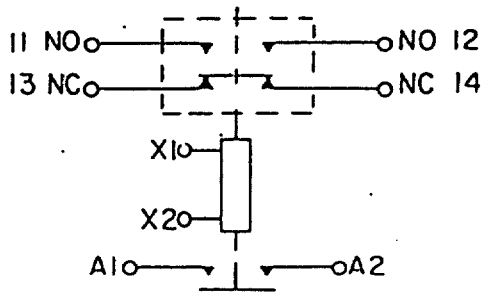
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D1 CIRCUIT



D2 CIRCUIT



PREPARING ACTIVITY: Air Force - 85 CUSTODIANS: ARMY - ER NAVY - AS AIR FORCE - 85 DLA - REVIEW: USER: PROJECT NUMBER: 5945-0840	MILITARY SPECIFICATION SHEET TITLE RELAY, 300 AMPERES, 1PST, N.O. TYPE I, HERMETICALLY SEALED	SPECIFICATION SHEET NUMBER **MS24184K** 26 July 1990 SUPERSEDING MS24184J 24 September 1982 AMSC N/A FSC 5945
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Inches	mm	Inches	mm	Inches	mm	Inches	mm
.010	0.25	.312	7.92	1.172	29.77	1.875	47.63
.031	0.79	.328	8.22	1.380	35.05	2.062	52.37
.062	1.57	.375	9.53	1.437	36.50	2.815	71.50
.063	1.60	.453	11.51	1.500	38.10	3.000	76.20
.091	2.31	.719	18.26	1.630	41.40	3.315	84.20
.138	3.51	.750	19.05	1.820	46.23	3.640	92.46
.240	6.10	.922	23.42	1.844	46.84	3.700	93.98
.266	6.76	.969	24.61				

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.031.
4. Coil and auxiliary terminals may use additional washer for terminal seat.
5. Weights include covers and barriers.
6. All test requirements of specification MIL-R-6106 apply except off time for motor load shall be six seconds minimum and average contact voltage drop reading before tests shall not exceed .15 volt.
7. PIN MS24184-1 inactive for design after 10 December 1957.
   PIN MS24184-D1 replaces MS24184-1.
8. Referenced documents shall be of the issue in effect on date of invitation for bid.

TABLE I. PIN and characteristics.

PIN MS24184- 1/ 2/	Type	Coil type	Terminal type	Mounting or mating socket	Auxiliary contacts	Max weight in pounds
D1	I	dc	Stud	Flange	N/A	2.4
D2	I	dc	Stud	Flange	With	2.5

1/ MS24184-A1 has been canceled without replacement.
2/ The term Part or Identifying Number (PIN) is equivalent to MS dash numbers.

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PREPARING ACTIVITY: Air Force - 85
CUSTODIANS: ARMY - ER NAVY - AS AIR FORCE - 85 DLA -
REVIEW:
USER:
PROJECT NUMBER: 5945-0840

MILITARY SPECIFICATION SHEET
TITLE
RELAY, 300 AMPERES, 1PST, N.O. TYPE I, HERMETICALLY SEALED

SPECIFICATION SHEET NUMBER
MS24184K
SUPERSEDING
MS24184J 24 September 1982
AMSC N/A FSC 5945

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TABLE II. Operating characteristics.

PIN MS24184-	Coil data									Time milliseconds max					
	Coil	Nominal			Max		Max pick-up voltage		Dropout voltage	Hold voltage	Operate	Release	Contact bounce 3/		
		Volts 1/	Freq. Hz	Res. +15% -10	Volts	Amp.	Normal 2/	High temp test	Cont. current test	2/	2/	4/	Main NO	Main NC	Aux NO	Aux NC
D1	X_1, X_2	28	dc	52	29	.6	18	21	22.5	1.5	7	35	15	2.5	---	---
D2	X_1, X_2	28	dc	52	29	.6	18	21	22.5	1.5	7	35	15	2.5	---	5	5

1/ Caution: use of any coil voltage less than the nominal coil voltage will compromise the operation of the relay.
2/ Over temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.

PREPARING ACTIVITY: Air Force - 85 CUSTODIANS: ARMY - ER NAVY - AS AIR FORCE - 85 DLA - REVIEW: USER: PROJECT NUMBER: 5945-0840	MILITARY SPECIFICATION SHEET TITLE RELAY, 300 AMPERES, 1PST, N.O. TYPE I, HERMETICALLY SEALED	SPECIFICATION SHEET NUMBER MS24184K SUPERSEDING MS24184J 24 September 1982 AMSC N/A FSC 5945	
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TABLE III. Rated contact load (amperes per pole) case grounded.

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				See appropriate notes
		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	50	300		5	5	300		5		
Inductive	50	100		5	5			5		
Motor 1/	50	250				150				
Lamp				.75	.75			.75		
Transfer load										2/
Mechanical life reduced current	100	75		1.25	1.25	75		1.25		
Intermediate current	50	AS APPLICABLE PER MIL-R-6106								

1/ Off time for motor load shall be 6 seconds min.
2/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.

ENVIRONMENTAL CHARACTERISTICS

Temperature rating:	-70°C to +71°C
Maximum altitude rating:	80,000 feet
Shock g-level:	25 g
Duration	6-9 ms
Max duration contact opening	2 ms
Vibration-sinusoidal (per table IV)	
G-level (per table IV)	g
Frequency range	Hz
Vibration-random	
Application spec	N/A
Power spectral density	N/A
RMS g min	N/A
Frequency range	N/A
Curve	N/A
High shock	N/A
Acceleration	15 g

ELECTRICAL CHARACTERISTICS

Min insulation resistance, initial	100 MΩ
After life or environmental tests	50 MΩ
Dielectric strength (sea level)	2-5 s

	Initial		After life tests	
	28 V dc	115 V ac	28 V	115 V
Coil to case	1250	1500	1000	1125
Aux contacts	1250	1500	1000	1125
All other points	1250	1500	1000	1125

Dielectric strength (altitude) 1 minute

	28 V dc	115 V ac
Coil to case	500	500
Aux contacts	500	500
All other points	500	500

Average contact drop initial	.150 V
Max after life test	.175 V
Overload current (NO)	2400 A
Rupture current (NO)	3000 A
Duty rating	Continuous
RFI spec	MIL-STD-461

(Applicable to coil circuits of ac operated relays)

PREPARING ACTIVITY: Air Force - 85	MILITARY SPECIFICATION SHEET	SPECIFICATION SHEET NUMBER	
CUSTODIANS: ARMY - ER NAVY - AS AIR FORCE - 85 DLA -	TITLE: RELAY, 300 AMPERES, 1PST, N.O. TYPE I, HERMETICALLY SEALED	MS24184K	
REVIEW:		SUPERSEDING MS24184J	24 September 1982
USER:		AMSC N/A	FSC 5945
PROJECT NUMBER: 5945-0840			

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TABLE IV. Vibration level.

PIN MS24184-	Vibration level				
	5-10 Hz	10-55 Hz	55-250 Hz	250-500 Hz	500-1500 Hz
D1	.08 DA	.06 DA	10 g's	5 g's	4 g's
D2	.08 DA	.06 DA	10 g's	5 g's	4 g's

TABLE V. Qualification by similarity.

PIN MS24184-	Loads						Dynamics 1/			Environmental 1/		
	Type I			Type I ER								
	A	B	C	D	E	F	A	X	Y	A	B	C
D1	4						2			4		
D2	4 2/						2			4 2/		

1/ All relays must be tested. Reference MIL-R-6106, appendix I.
2/ Testing of relay with auxiliary contacts. Reference MIL-R-6106, appendix I.

Revision letters are not used to denote changes due to the extensiveness of the changes.

PREPARING ACTIVITY: Air Force - 85 CUSTODIANS: ARMY - ER NAVY - AS AIR FORCE - 85 DLA - REVIEW: USER: PROJECT NUMBER: 5945-0840	MILITARY SPECIFICATION SHEET TITLE RELAY, 300 AMPERES, 1PST, N.O. TYPE I, HERMETICALLY SEALED	SPECIFICATION SHEET NUMBER MS24184K SUPERSEDING MS24184J 24 September 1982 AMSC N/A FSC 5945

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